



LAB PROJECT

CSE350: Digital Electronics and Pulse Techniques

Project Title: Digital Voltmeter (Without Arduino and IC)

Section: 06-A

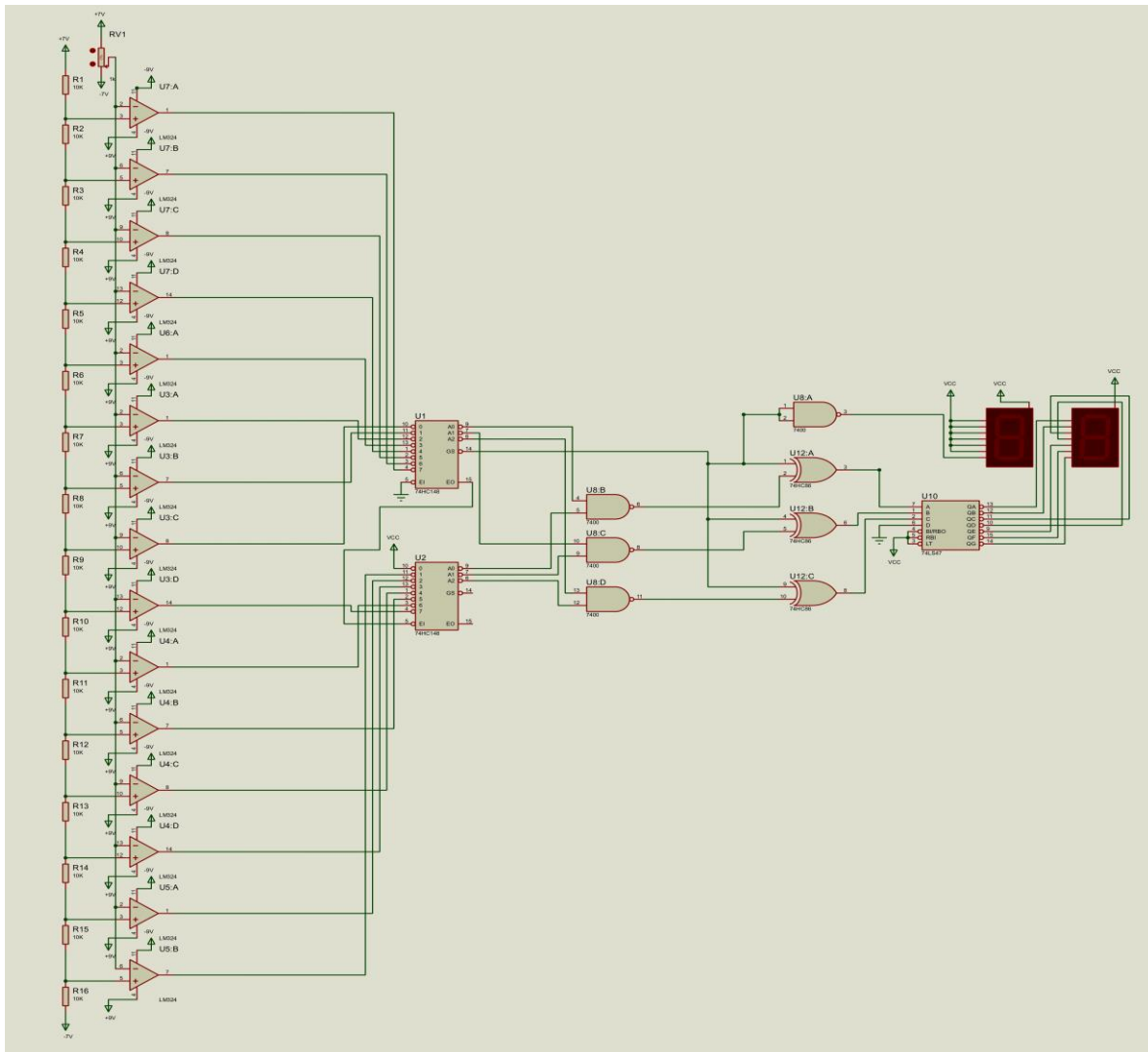
Group: 2

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Digital Voltmeter (Without Arduino and IC)

The Digital Voltmeter project is designed to measure analog voltage and present it on a digital display. The circuit is capable of detecting a range from -7V to +7V. Because the system utilizes a 4-bit Flash ADC, the hardware consists of a reference ladder with 16 resistors of 10Kohm and 15 op-amp comparators. The outputs of these op-amps are fed into two 8-to-3 bit active-low priority encoders to be converted into binary code. These signals are then processed through a logic block of NAND and XOR gates to extract the absolute magnitude of the voltage, which is then passed to a BCD decoder for the final display. The GS (Group Signal) pin of the primary encoder acts as the sign bit; it determines whether the voltage is positive or negative while the remaining three bits provide the magnitude for the numerical reading.

Circuit Diagram: [PDF](#)



Screenshot of the receipt of ordered components:



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IC - 74148 - Priority Encoder (SN74HC148) 8 to 3

₹105.00

~~₹75.00~~ ₹52.50

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10k Ohm 1/4W Resistor (Pack of 10)

₹40.00

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IC - LM324 - Quad Comparator Operational Amplifier (Op Amp) (LM324N)

₹270.00

~~₹30.00~~

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IC - 7400 - NAND Gate (SN74HC00N)

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Subtotal

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Workflow:

The process starts with the comparison of the analog input voltage with the 15 reference voltage levels established by a series-connected resistor ladder, causing the LM324 op-amps to generate an active-low thermometer code. This code is processed by two cascaded 74148 priority encoders that compress the signal into binary bits. The GS (Group Signal) from the primary encoder acts as a sign bit to distinguish between the positive and negative voltages. Then, the binary outputs are passed through 7400 NAND and 7486 XOR gates, which invert the bits only when a negative input voltage given. As a result, the display shows a positive magnitude for a negative input. Finally, this corrected 3-bit magnitude is sent to the 7447 BCD decoder, and the outputs of the decoder are connected with the specific segments of the 7-segment display, which shows both the sign and numerical value of the measured voltage.